

EB-JEPA — Reference Table (V1)

EB-JEPA EEG track

June 2026

Our EB-JEPA runs on TUAB (v3.0.1, patient-disjoint, 276 eval recordings). *per-rec* = per-recording (16-window mean-pool, the clinical metric); *per-win* = per-window (the literature convention). $\pm$ = 3-seed standard deviation; other rows are single-seed. Companion to `explanation_augmented_corruption.tex` (which details the augmentation) and `PROVENANCE_TABLE.md` (the literature baselines).

#	Encoder	Params	Energy (SSL objective)	Eval	per-rec BAcc / AUROC	per-win BAcc / AUROC
baseline	Conv1D	0.4M	VICReg — <i>base aug, no corruption</i>	frozen	0.796 / 0.888	0.756 / –
1	Conv1D	0.4M	VICReg + spectral 0.1	frozen	0.836 / 0.887	0.765 / –
2	Conv1D	0.4M	SIGReg	frozen	0.825 / 0.913	0.775 / 0.856
3	Conv1D	0.4M	VICReg	frozen	0.819 ± 0.004 / 0.900 ± 0.006	0.770 / 0.848
4	Conv1D	0.4M	VICReg	FT	0.812 ± 0.004 / 0.908 ± 0.006	–
5	Transformer (multi-corpus)	3.65M	VICReg	frozen	0.798 / 0.872	0.738 / –
6	Conv1D	0.4M	VICReg + spectral 0.1 + $\Delta z / \Delta z^2 / \text{ACF} / \text{sACF}$	frozen	0.823 ± 0.007 / 0.900 ± 0.010	0.761 ± 0.004 / 0.841 ± 0.004

Fixed for every corruption run (never swept): views v_1, v_2 made with Gaussian noise $\sigma = 0.1$, scale-jitter $\pm 20\%$, channel-drop $p = 0.2$, plus exact EB-corruption — 20% timepoints masked + 20% $\pm 6\sigma$ outliers ($\approx 40\%$ /view). The **baseline** is the no-corruption reference (legacy contiguous time-mask).

Row 6 — stylized-facts energy (3-seed). Row 1’s energy plus four view-consistency blocks between the two views’ encoder feature maps, each an MSE between a path-statistic of v_1 and v_2 , with explicit, fixed a-priori (not eval-swept) coefficients: increments Δz (0.10), realized variance $E[\Delta z^2]$ (0.05), temporal autocorrelation over lags 1–8 (0.10), and autocorrelation of the power spectrum / FFT (0.05); the spectral term stays at its validated 0.10. All ≤ 0.1 so VICReg’s variance/covariance anti-collapse terms remain dominant. *Result: squarely on the 0.82 plateau* — the added path-statistics regularise the representation but do not lift the 0.4M-conv ceiling (per-window is, if anything, ~ 1 pp lower than row 3).